

- 8 (a) Non-cyclic photophosphorylation occurs in the light-dependent stage of photosynthesis.

Outline the main features of non-cyclic photophosphorylation.

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..... [4]

- (b) Some Calvin cycle intermediates are used to produce other molecules that are **not** part of the Calvin cycle.

Name **two** of these intermediates.

For each intermediate, state **one** example of the type of molecule that is produced.

intermediate

molecule produced

intermediate

molecule produced

[2]

- (c) Fig. 8.1 shows how the rate of uptake of carbon dioxide by a plant varies with temperature, at an optimum intensity of light on a clear day.

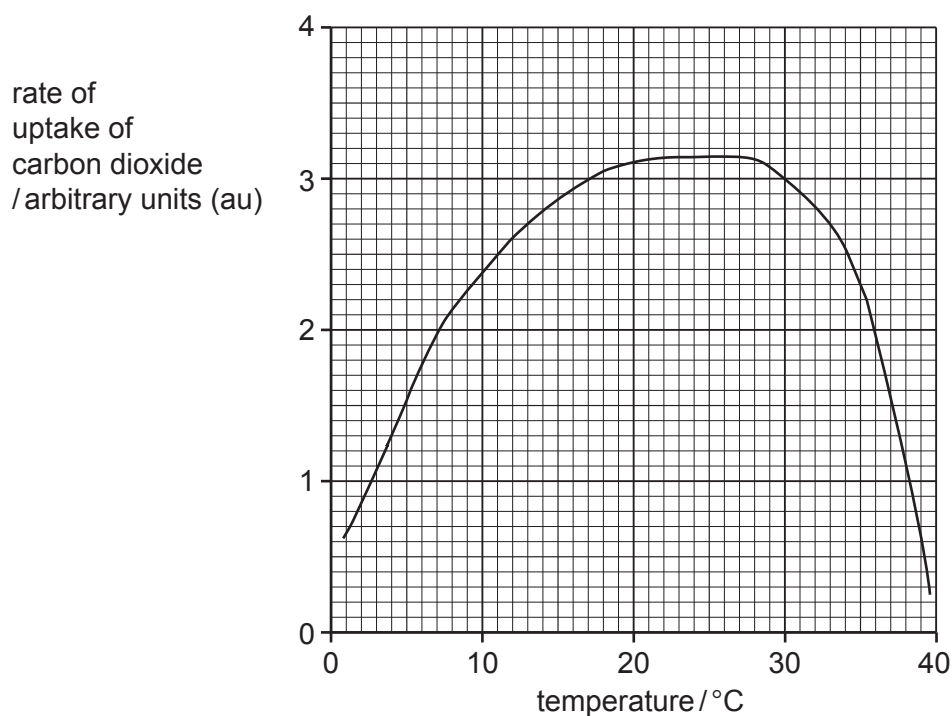


Fig. 8.1

- (i) Calculate the mean increase in rate of uptake of carbon dioxide as the temperature increases from 5°C to 20°C.

Show your working.

Give your answer to **two** decimal places.

..... au °C⁻¹ [2]

- (ii) Suggest why the rate of carbon dioxide uptake levels off and then decreases after 20°C.

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..... [3]

- (iii) On Fig. 8.1, draw a curve to show the uptake of carbon dioxide on a cloudy day. [1]

[Total: 12]